INNOMATICS RESEARCH LABS

Data-Driven Insights: Flipkart Comic & Manga Markets (India • Japan • US)

By: Sanket Jadhav

Exploratory Data Analysis on Flipkart Comic Books Dataset

About Me

- My name is Sanket Jadhav, a recent graduate in Bachelor of Computer Applications(BCA).
- I am passionate about using data to uncover patterns, solve problems, and support smarter decision-making

Profiles

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Problem Statement:

As a student who enjoys Marvel, DC and anime, I want to buy my first comic or manga. But I have limited pocket money. So what country, publisher or writer offers the best value-for-money comic—one that is affordable, highly rated, and loved by readers?

Market questions answered by this EDA :

Which country's comics cost more?

Which categories earn better ratings?

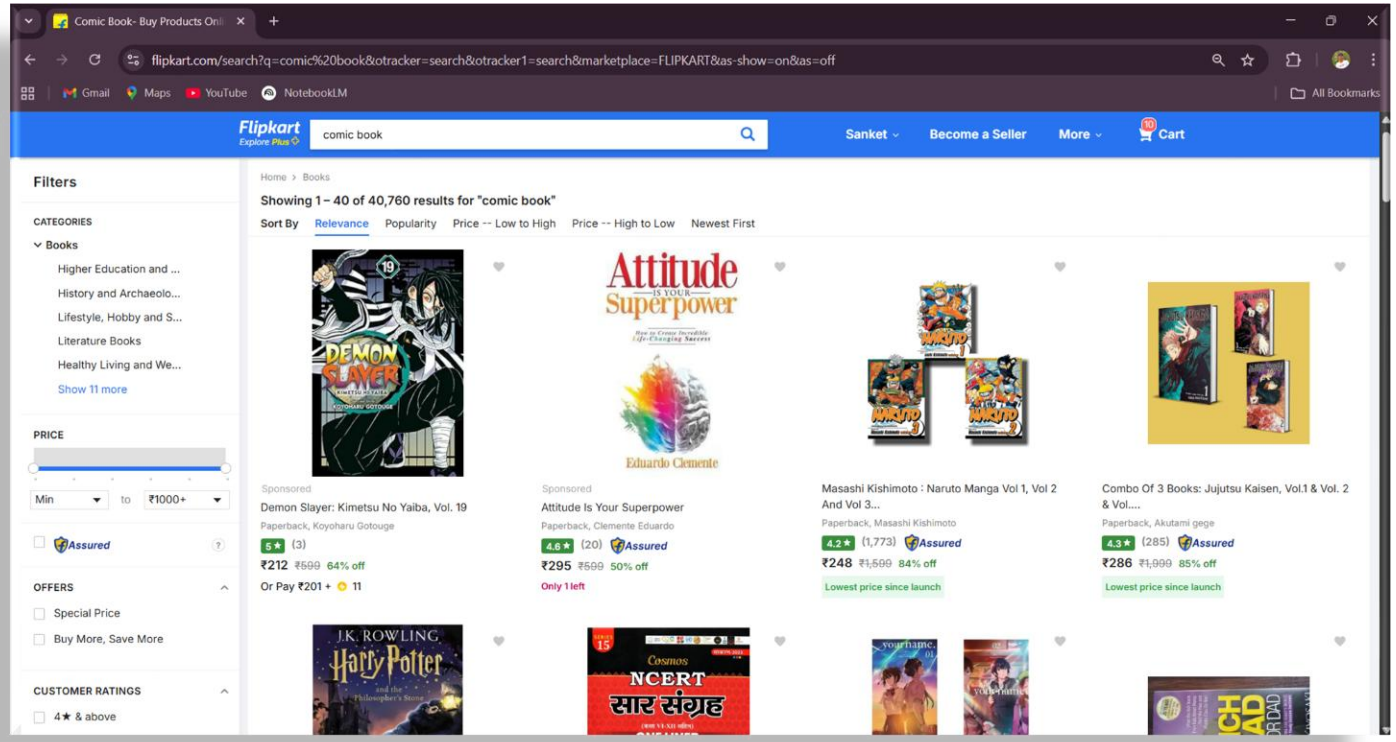
Who are the dominant publishers in India?

Outcome: This analysis helped identify which comic or manga provides the best value for money for a student. By comparing prices, discounts, ratings, reviews, formats, and origins (India, Japan, US), the study reveals where high-quality and budget-friendly comics are available and which publishers and writers are most suitable for beginners.

Requests:
Retrieve HTML pages via HTTP using Requests for web access

BeautifulSoup:
Parse HTML to locate elements: Title, Author/Publisher, Genre, Format

Regex cleaning :
Extract and normalize Price, Discount, Rating, Number of Reviews using Regex



Pandas structuring:
Assemble cleaned fields into a DataFrame for analysis and export

Final dataset:
Columns: Title, Country, Price, Discount, Rating, Number of Reviews, Author/Publisher, Genre, Format

Data Cleaning — Before & After

Key cleanup actions that improve analysis accuracy

Column	Before	After
MRP	₹1,758	1758
Discount	48% off	48.0
Product Review	(3)	3
Duplicates	multiple identical rows	duplicates removed
Missing values	empty fields present	handled / imputed

Initial Scraped Dataset Overview:

Ready for analysis

Categorical Values

	product_price	product_rating	mrp	company	country	discount_pct	review_count	availability_status	language	writer	format	comic title
0	899	2.3	1758.0	Marvel	United States	48.0	3	low_stock	English	Hickman Jonathan	paperback	X-Men: Hellfire Gala
1	1178	0.0	1316.0	Marvel	United States	10.0	0	low_stock	English	Dbg	paperback	World of Reading Avengers Boxed Set
2	617	0.0	822.0	Marvel	United States	24.0	0	low_stock	English	Berrios Frank	hardcover	Trapped by the Green Goblin! (Marvel: Spider-Man)
3	522	3.9	699.0	Marvel	United States	25.0	8	low_stock	English	Dk	hardcover	Marvel Avengers The Ultimate Character Guide N...
4	899	5.0	1949.0	Marvel	United States	53.0	3	low_stock	English	Aaron Jason	paperback	Avengers By Jason Aaron Vol. 9
...	...	...	...	...	...	...	...	...	...	...	...	...
884	591	3.3	969.0	Japanese	Japan	39.0	3	promotion	English	Kishimoto Masashi	paperback	Boruto: Naruto Next Generations, Vol. 15
885	591	4.4	799.0	Japanese	Japan	26.0	13	in_stock	English	Kishimoto Masashi	paperback	Naruto, Vol. 55
886	591	4.8	799.0	Japanese	Japan	26.0	4	in_stock	English	Kishimoto Masashi	paperback	Naruto, Vol. 13
887	866	4.4	1655.0	Japanese	Japan	47.0	12	promotion	English	Aidairo Aidairo	paperback	Toilet-bound Hanako-kun, Vol. 12

Numerical Values

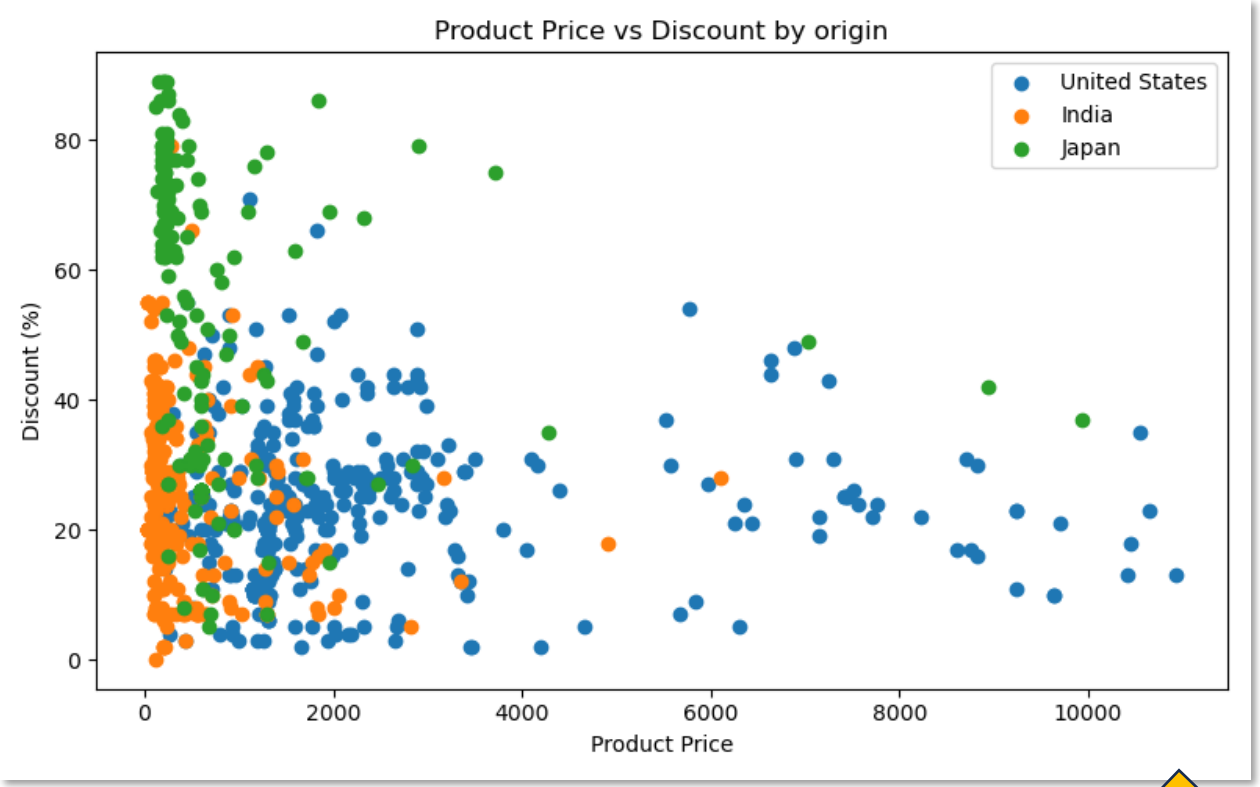
“Where Should I Look First? (Price + Discount Comparison)”

Insights :

1. **Japan offers the highest discounts**
   - **higher discount percentages**.

2. **India is the most budget-friendly overall**
   - Moderate discounts + very **affordable** prices = strong value.

3. **United States comics are the most expensive**
   - US products cluster on the **higher end of the price range**.
   - Feels like a “premium market,” not suitable for limited pocket money.



Scatter plot comparing Product Price vs Discount across India, Japan, and the US

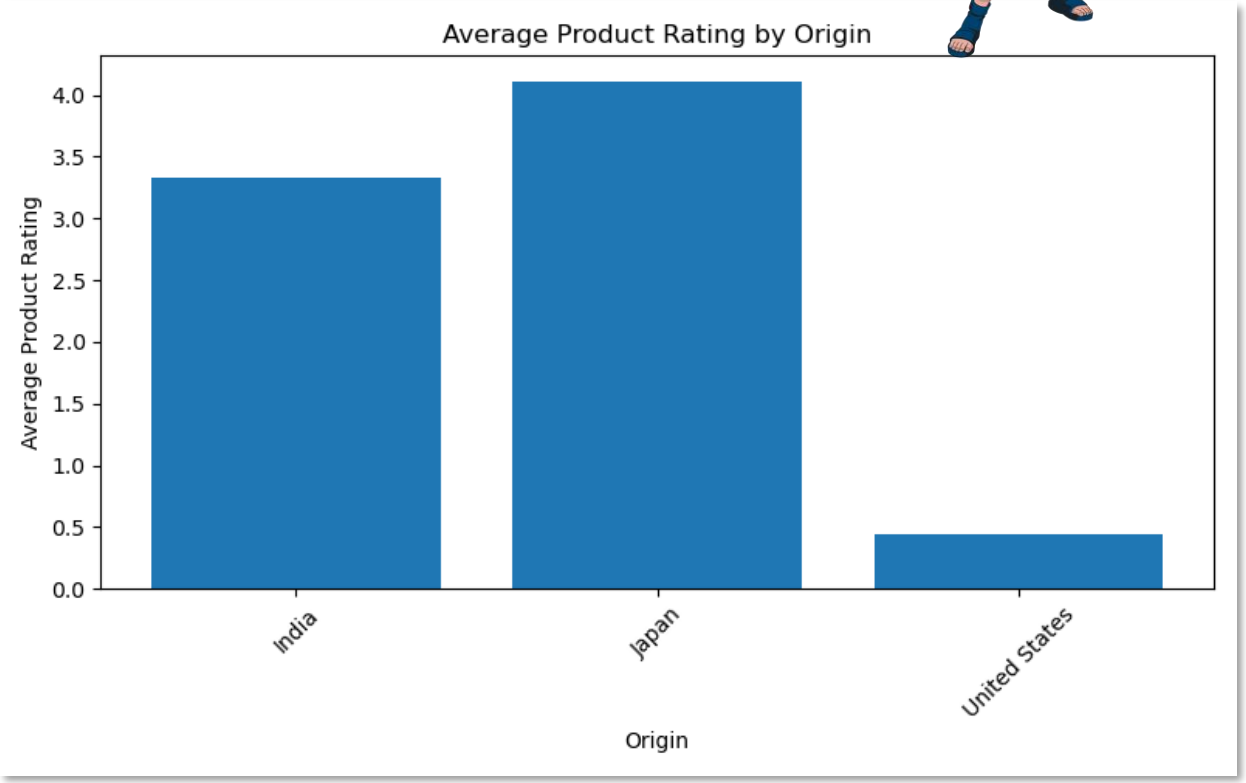
Conclusion:

India and Japan offer the best starting point for a beginner comic reader looking for value-for-money.

Does High Discount Mean Low Quality?

Insights :

1. **Japan has the highest average rating (4.1★)**
- Strong signal of **quality storytelling and strong fan engagement**.
2. **India has decent ratings (3.3★)**
- Good balance of **affordability + quality**.
3. **United States has surprisingly low average rating (0.45★)**
   Higher price doesn't always mean higher audience satisfaction



Average Product Rating by Origin (India, Japan, US)

Conclusion:

Japan leads in quality, India is a good balance, and US comics show poor engagement — making Japan and India the best choices for a beginner like me.

What Do Other Readers Think?

Insights:

1. **Japan has the widest range of review counts.**
- Some Japanese comics reach **600–800 reviews**, showing very high reader engagement.

2. India shows moderate review activity with many mid-range counts.
- A lot of Indian comics fall between **20–150 reviews**, showing consistent but not extreme engagement.

3. United States has very low review counts on Flipkart.
- Most US titles have **0–10 reviews**, indicating weak platform engagement even though globally they are popular.

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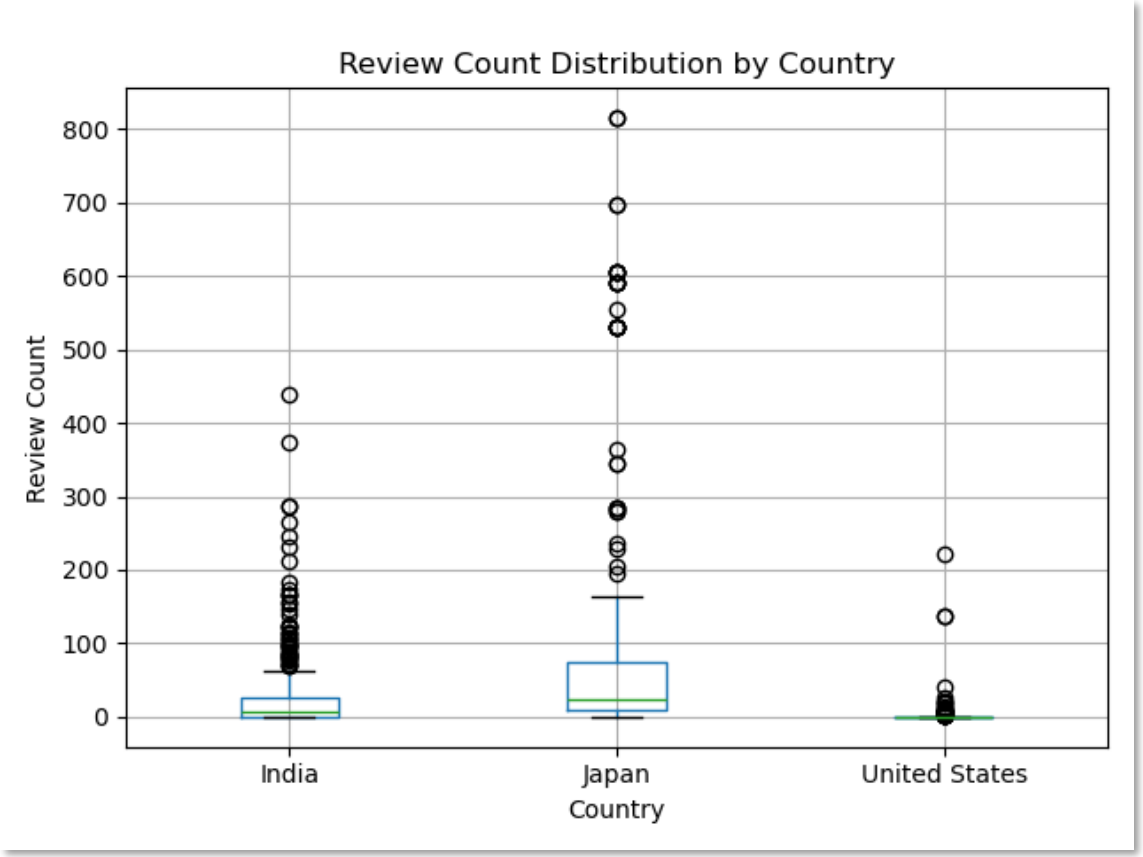


Review Count Distribution by Country

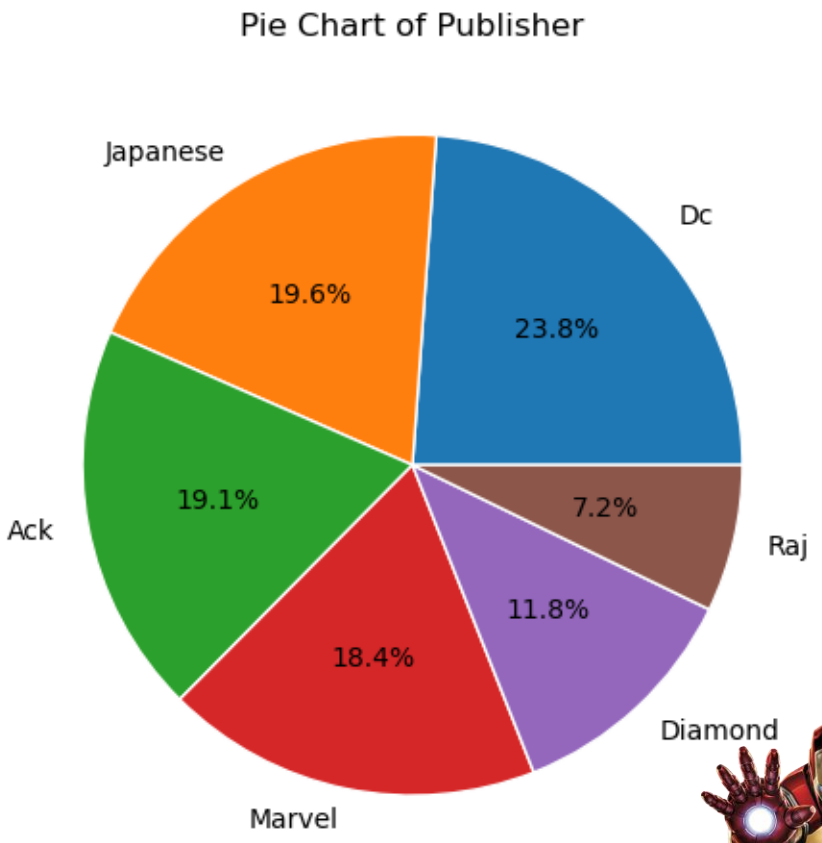
Conclusion:

Japan leads in reader engagement, India follows with consistent mid-level activity, and the US has very low participation.

Which Publishers Dominate the Market?

Publisher Preferences in Comics

- **DC Comics** leads with 23.8%, indicating a strong presence and possibly a wider range of titles or popularity among readers.
- **Japanese (19.6%), ACK (19.1%),** and **Marvel (18.4**%) follow closely, showing competitive representation and diverse content offerings**.**
- **Diamond (11.8%)** and **Raj (7.2%)** make up smaller segments, likely catering to niche audiences or specific regional markets.



Pie Chart of Publisher Distribution

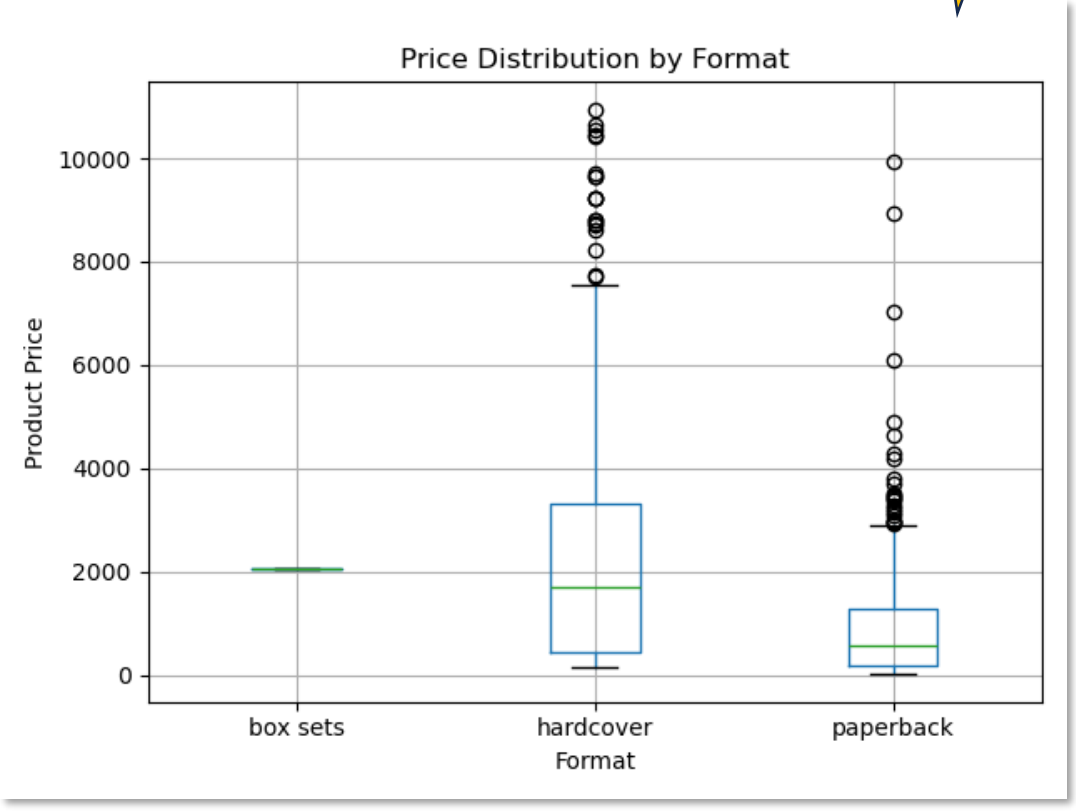
Conclusion:

DC, Japanese publishers, and ACK provide the widest variety of comic options, making them ideal to start exploring based on personal preference and budget.

Which Format is Best?

Insights :

1. **Hardcovers are the most expensive format**
   - Prices reach very high levels, making them unsuitable for students.
2. **Paperbacks are the most affordable**
   - Stable price range with very low median cost
   - Best format if you want to explore comics cheaply
3. **Box sets are consistently priced but not budget-friendly**
   - Good only if someone wants a bundled experience



Price Distribution by Format
(Box sets, Hardcover, Paperback)

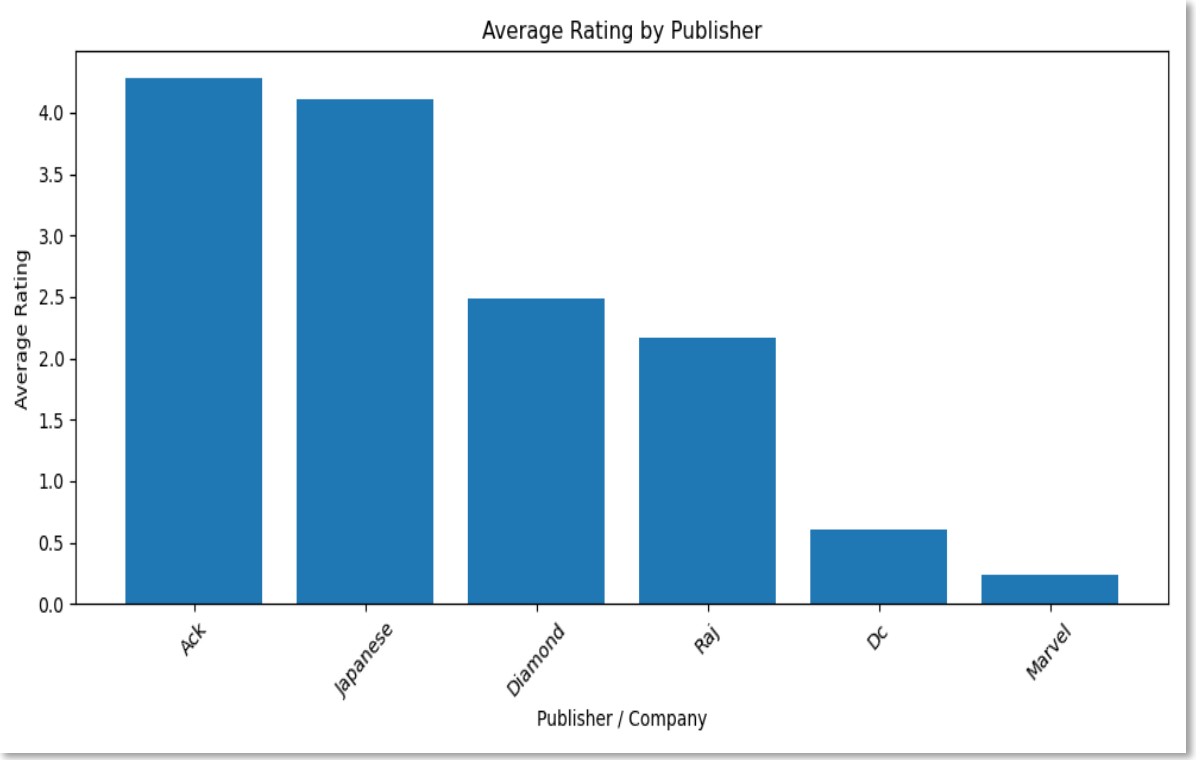
Conclusion :

Paperback is the most budget-friendly and beginner-friendly choice, offering the best balance between cost and content.

Which Publisher Should I Trust?

Insights:

1. ACK and **Japanese** publishers have the highest ratings **(4.1–4.3★).**
2. DC and **Marvel** show surprisingly **low ratings** on Flipkart. Despite high global popularity, their listed products receive lower audience satisfaction here.



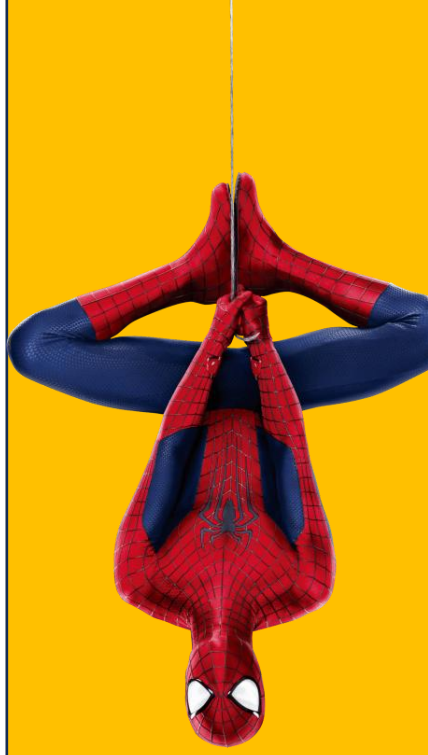
Average Rating by Publisher

Conclusion :

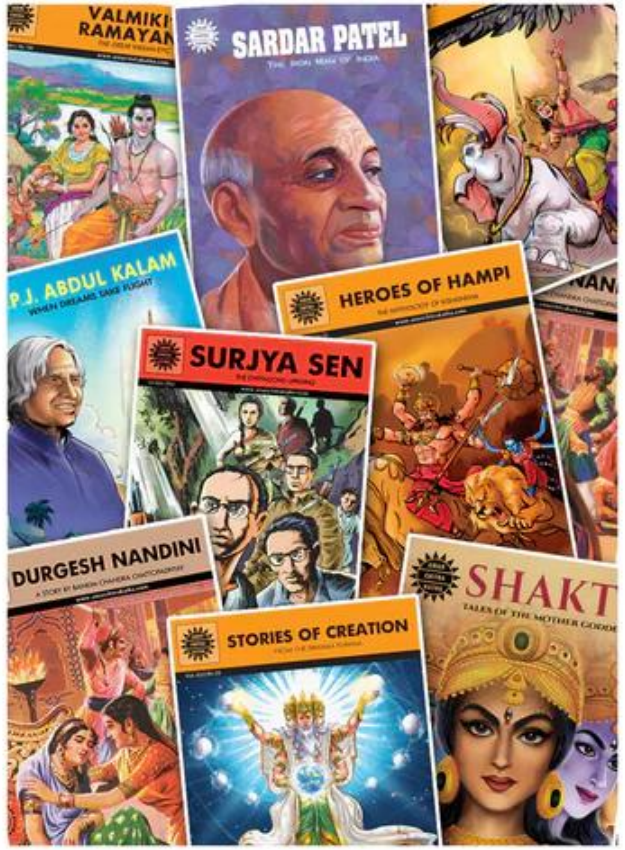
Publisher quality matters for a first-time reader.
Based on the ratings, ACK and Japanese publishers provide the best reader satisfaction, making them ideal starting points for someone who wants both quality storytelling and a trustworthy reading experience.

Conclusion :

1. Japan consistently provides the best combination of high discounts, strong ratings and high reader engagement.
2. India offers the most affordable prices and a stable mid-level quality, making it a safe budget-friendly option.
3. US comics, although globally famous, are the most expensive and show very low interaction on Flipkart.
4. Paperback format delivers the best value-for-money, while hardcovers are too costly for beginners.
5. Publishers like ACK and Japanese brands give the highest reader satisfaction, making them ideal for first-time readers.



So… Which Comic Did I Finally Choose?



ACK (India)

affordable, high ratings, beginner-friendly

Japanese Manga

high discounts, best ratings, and massive global popularity

ONE PIECE
尾田栄一郎
鉄腕アトム
ATOM
JUMP COMICS
鬼滅の刃
吾峠呼世晴
DRAGON BALL
鳥山明
KATSUHIRO
SPY×FAMILY
遠藤達哉
JUMP COMICS
らんま1/2
Shonen Sunday

THANK YOU